

# D-RDW: Diversity-Driven Random Walks for News Recommender Systems

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## Abstract

This paper introduces Diversity-Driven Random Walks (D-RDW), a lightweight algorithm and re-ranking technique that generates diverse news recommendations. D-RDW is a societal recommender, which combines the diversification capabilities of the traditional random walk algorithms with customizable target distributions of news article properties. In doing so, our model provides a transparent approach for editors to incorporate norms and values into the recommendation process. D-RDW shows enhanced performance across key diversity metrics that consider the articles' sentiment and political party mentions when compared to state-of-the-art neural models. Furthermore, D-RDW proves to be more computationally efficient than existing approaches.

## CCS Concepts

• **Information systems** → **Information retrieval diversity; Recommender systems**; • **Theory of computation** → **Random walks and Markov chains**.

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## 1 Introduction

In news recommender systems (NRSs), diversity is one of the most important criteria for assessing the societal impact of these systems [6, 38]. This is particularly true for recommending political articles, as a diverse selection of news can impact opinion formation and social deliberation [17–19]. When designing NRSs, however, diversity optimization is predominantly achieved in a post-processing step [29, 36]. As such, it plays only a secondary role; it is limited to re-ranking items from the candidate list of the model, instead of being able to access the entire item pool [41]. To address the limitations of re-ranking approaches, this paper introduces *Diversity-Driven Random Walks* (D-RDW) that provides diversity optimization at the model stage.

D-RDW is an extension of the random walk algorithms  $RP_\beta^3$  [11, 27]. In the past,  $RP_\beta^3$  was shown to outperform neural models across a wide range of accuracy and diversity metrics in the movie and e-commerce domain [11, 15]. While state-of-the-art neural models

can account for diversity through re-ranking, the benefits of using random walks for news recommendations are that they allow for 1) an explainable recommendation approach (i.e., they are not a black box model), 2) an easily extendable approach to include heuristics of editors and journalists, and 3) a lightweight and cost-efficient approach that is highly scalable and parallelizable. D-RDW achieves this by introducing a new step in the random walk pipeline that enforces a normative target distribution (NTD)<sup>1</sup> of article properties in the recommendation list. As a secondary contribution, we demonstrate the effectiveness of enforcing NTD to achieve higher diversity by creating separate re-ranking strategies for existing neural models. In doing so, we offer a first diversity-optimizing model and re-ranker for a *deliberative recommender* according to the normative NRSs framework of Helberger [19].

## 2 Related Work

Diversity has increasingly been recognized as a crucial aspect of assessing the quality of news recommendations [5], by looking at aspects such as topics, viewpoints, and political mentions, among others [2, 3, 6, 22, 31]. NRSs also take into account other forms of diversity, including those related to sources, people, events, semantics, sentiments, authors, genders, and temporal aspects [10, 45]. In recent years, however, the research community has begun to recognize the critical role that NRSs play in democratic societies and proposed four different models from the normative framework of democracy to consider exposure diversity as a societal goal [19, 20]. These normative frameworks, however, have not yet been implemented into any recommendation model.

Looking at models, several approaches in the literature focused on generating such diversified recommendations by using random walk-based models, i.e., models that simulate user navigation in an item-user interaction graph as a probabilistic transition [4, 13, 14, 26, 42, 47]. Liu et al. [25] introduced a collaborative filtering method based on directed random walks that amplifies the influence of users with small degrees to reach a more diverse set of recommendations in the movie domain. Christoffel et al. [11, 28] created an improved variant,  $RP_\beta^3$ , that showed the capability of random walks to optimize for both accuracy as well as diversity. Paudel and Bernstein [27] then proposed Random Walk with Erasure (RWE-D), where they successfully leveraged random walks to diversify recommendations within a left-right political spectrum.

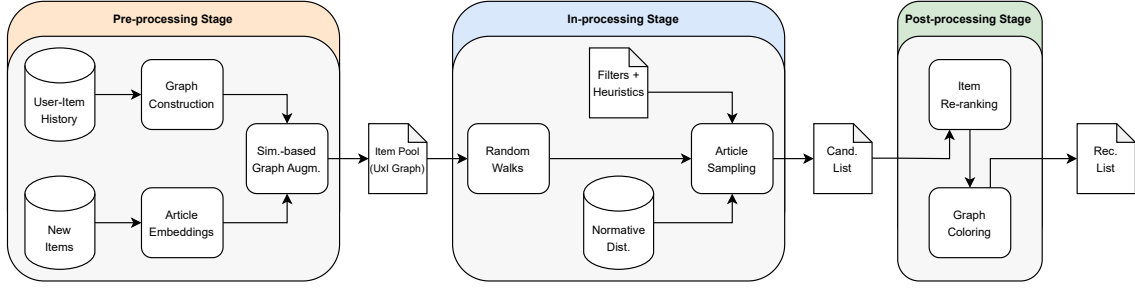
These approaches, however, do not offer any direct control to editors or journalists when it comes to specifying the precise dimension of diversity (be that, e.g., source or viewpoint diversity). The underlying algorithm only looks at user interactions and has no information on the article content. D-RDW addresses this issue with NTD-based filtering by accessing annotated news articles.

<sup>1</sup>When speaking of normativity in NRS, we follow the definition of Vrijenhoek et al. [40] which refers to the process of operationalizing and including societal values in the optimization and evaluation of NRS (e.g., optimizing news feeds for source or political viewpoint diversity and providing equal exposure for minority and majority parties).

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**Figure 1: The D-RDW pipeline consists of three stages: 1) a pre-processing stage for graph construction, 2) an in-processing stage for generating a candidate list, and 3) a post-processing step for creating the recommendation list.**

### 3 Diversity-Driven Random Walks

D-RDW follows the recommendation pipeline shown in Figure 1. It consists of three main steps: 1) a pre-processing stage for graph construction based on user-item interactions, 2) an in-processing stage for running random walks and applying filter criteria to enforce a target normative item distribution, and 3) a post-processing stage for re-ranking the candidate list to achieve a homogeneous distribution of articles across the news feed.<sup>2</sup>

**Stage 1: Graph Construction and Augmentation:** The random walk algorithm uses a bipartite graph with user and item nodes [11, 27], with edges representing user-item interactions (i.e., clicks). To address the cold start problem, unconnected nodes (i.e., articles not yet read by any user) are integrated by 1) selecting the most similar items already in the graph<sup>3</sup> and 2) creating edges between the new item and *all users* connected to its most similar counterpart already part of the graph (we connect to the 3 most similar items).

**Stage 2: Random Walks and Normative Distributions.** Recommendations are generated by following four steps: 1) running random walks on the graph with 3 hops, 2) a heuristic rule for removing items included in the history of the target user, 3) applying a norm-aware target distribution (NTD),<sup>4</sup> and 4) sampling random items from test impressions to fill open positions. The novel element of D-RDW is the article sampling in Algorithm 1 on Line 7 to solve the constraint satisfaction problem of the NTD. For example, let  $I$  be an all-ones vector in a matrix form  $I^T X = 20$ , where  $X$  is the target size of the news feed of 20 items. We then define a binary indicator matrix for each category. Take  $P$ , where  $P_i = 1$  represents the political item  $i$  and let  $S_i = 1$  be an item  $i$  with a positive sentiment. In this example, our NTD should include 15 political articles and 10 articles with a positive sentiment. By combining these linear equations into their matrix form, we arrive at Equation 1:

$$\begin{pmatrix} I^T \\ P^T \\ S^T \end{pmatrix} X = \begin{pmatrix} 20 \\ 15 \\ 10 \end{pmatrix} \quad (1)$$

<sup>2</sup>A reference implementation of our algorithm is available online in our repository: <https://anonymous.4open.science/t/AnonymousExperimentsRepository-C04E>

<sup>3</sup>Items are matched based on the semantic similarity of the article text. Similarity is calculated with an off-the-shelf model: <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

<sup>4</sup>We provide a norm-aware deliberative recommender [19] by focusing on article sentiment and political party. However, this is one possible example, as deliberative recommenders can be operationalized in different ways (see Section 4 for details).

The variable  $X$  is restricted to binary values (0 or 1), making it a binary integer programming problem. As multiple solutions for  $X$  may satisfy these linear equations, we use recency as an objective function. Let  $C$  represent a vector of item attributes that the sampling algorithm aims to maximize in the final item selection. When  $C$  is defined as the item recency vector, the sampling goal is to select items that satisfy the linear constraints while maximizing overall recency, formalized as  $\max \sum_i c_i \cdot x_i$ .<sup>5</sup>

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#### Algorithm 1: Diversity-Driven Random Walks

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**Input :** *graph, articleAttributes, targetDimensions, targetDistribution, feedSize, filterCriteria, samplingObjective, maxHops*

**Output:** *recommendations*

```

1 recommendations = []
2 currentHop = 3
3 while True do
4   candidates = newHop(graph, currentHop)
5   candidates = filterHeuristics(candidates,
6     articleAttributes, filteringCriteria)
7   recommendations = sampling(candidates,
8     articleAttributes, targetDimension,
9     targetDistribution, targetSize,
10    samplingObjective)
11   if length(recommendations) >= targetSize ∨
12     currentHop >= maxHops then
13     break
14   currentHop = currentHop + 2
15 if length(recommendations) < targetSize then
16   recommendations = addRandomArticles(candidates,
17     articleAttributes, targetDimensions,
18     targetDistribution, targetSize,
19     samplingObjective)
20 recommendations = rankArticles(recommendations,
21   targetSize, rankingObjectives)
22 return recommendations

```

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<sup>5</sup>News items have a short shelf life, high turnover, and short consumption time. By prioritizing recency, we can ensure users only see the most up-to-date stories on a given topic, keeping them engaged for longer.

**Table 1: Overview of the diversity scores for the top 20 news recommendations and AUC scores for predicting user impressions. The values closest to the perfect score are highlighted in green, second closest in blue, and third closest in red.**

| Model               | Re-ranking | Activ. | Cat. Calib. | Comp. Calib. | Frag. | Alt. Voices | Repr. | Cat. Gini | Sent. Gini | Party Gini | Cat. ILD | Sent. ILD | Party ILD | Train. Cost | Rec. Cost | AUC   |
|---------------------|------------|--------|-------------|--------------|-------|-------------|-------|-----------|------------|------------|----------|-----------|-----------|-------------|-----------|-------|
| Perfect (NTD)       | 0.000      | 1.000  | 0.000       | 0.000        | 0.000 | 0.000       | 1.000 | 0.000     | 0.133      | 0.250      | 1.000    | 0.779     | 0.789     | N/A         | N/A       | 1.000 |
| LSTUR [1]           | N/A        | 0.190  | 0.433       | 0.257        | 0.610 | 0.063       | 0.372 | 0.832     | 0.645      | 0.874      | 0.740    | 0.552     | 0.343     | 351.12      | 69.84     | 0.564 |
| NPA [43]            | N/A        | 0.202  | 0.468       | 0.266        | 0.619 | 0.061       | 0.365 | 0.826     | 0.605      | 0.877      | 0.753    | 0.580     | 0.337     | 523.05      | 51.25     | 0.554 |
| NRMS [44]           | N/A        | 0.204  | 0.509       | 0.285        | 0.632 | 0.060       | 0.362 | 0.791     | 0.544      | 0.880      | 0.794    | 0.624     | 0.326     | 326.55      | 63.23     | 0.549 |
| LSTUR + NTD         | G-KL       | 0.282  | 0.444       | 0.240        | 0.612 | 0.104       | 0.546 | 0.828     | 0.133      | 0.250      | 0.754    | 0.779     | 0.789     | 351.12      | 30.15     | N/A   |
|                     | PM-2       | 0.284  | 0.440       | 0.244        | 0.604 | 0.115       | 0.546 | 0.819     | 0.150      | 0.250      | 0.766    | 0.776     | 0.789     | 351.12      | 23.86     | N/A   |
|                     | MMR        | 0.295  | 0.433       | 0.235        | 0.613 | 0.118       | 0.575 | 0.823     | 0.226      | 0.270      | 0.759    | 0.762     | 0.807     | 351.12      | 52.36     | N/A   |
| NPA + NTD           | G-KL       | 0.292  | 0.469       | 0.237        | 0.589 | 0.108       | 0.545 | 0.810     | 0.133      | 0.250      | 0.775    | 0.779     | 0.789     | 523.05      | 30.15     | N/A   |
|                     | PM-2       | 0.302  | 0.471       | 0.239        | 0.576 | 0.122       | 0.547 | 0.806     | 0.150      | 0.250      | 0.783    | 0.776     | 0.789     | 523.05      | 23.86     | N/A   |
|                     | MMR        | 0.314  | 0.463       | 0.236        | 0.598 | 0.110       | 0.575 | 0.808     | 0.219      | 0.270      | 0.772    | 0.763     | 0.807     | 523.05      | 52.36     | N/A   |
| NRMS + NTD          | G-KL       | 0.307  | 0.482       | 0.239        | 0.608 | 0.091       | 0.544 | 0.798     | 0.133      | 0.250      | 0.790    | 0.779     | 0.789     | 326.55      | 30.15     | N/A   |
|                     | PM-2       | 0.316  | 0.483       | 0.244        | 0.598 | 0.099       | 0.544 | 0.796     | 0.150      | 0.250      | 0.794    | 0.776     | 0.789     | 326.55      | 23.86     | N/A   |
|                     | MMR        | 0.315  | 0.477       | 0.238        | 0.616 | 0.099       | 0.571 | 0.798     | 0.204      | 0.270      | 0.790    | 0.767     | 0.807     | 326.55      | 52.36     | N/A   |
| D-RDW               | N/A        | 0.374  | 0.407       | 0.229        | 0.394 | 0.107       | 0.556 | 0.798     | 0.133      | 0.250      | 0.810    | 0.779     | 0.789     | 0.79        | 4.12      | 0.554 |
| $RP_{\beta}^3$ [11] | N/A        | 0.222  | 0.415       | 0.235        | 0.582 | 0.080       | 0.376 | 0.840     | 0.755      | 0.856      | 0.743    | 0.439     | 0.392     | 0.54        | 0.40      | 0.565 |
| RWE-D [27]          | N/A        | 0.256  | 0.435       | 0.222        | 0.443 | 0.100       | 0.372 | 0.857     | 0.802      | 0.842      | 0.735    | 0.377     | 0.433     | 0.61        | 0.08      | 0.554 |
| Random              | N/A        | 0.180  | 0.461       | 0.256        | 0.705 | 0.054       | 0.366 | 0.756     | 0.634      | 0.873      | 0.842    | 0.564     | 0.346     | N/A         | N/A       | 0.500 |

*Stage 3: Graph Coloring and Item Re-ranking.* D-RDW sorts the candidate list by the prediction score in descending order and applies a graph coloring algorithm [23] to the candidate list (where the article categories represent colors). This ensures a homogeneous category distribution across the entire recommendation list. Line 15 performs random sampling from the most recent news items to complete the targeted size of the news feed.

## 4 Experiment

We evaluate D-RDW on the RecSys 2024 Challenge benchmarking Ekstra Bladet News Recommendation Dataset (EB-NeRD), specifically EB-NeRD small [24]. In our analysis, we compare the performance of D-RDW with the neural models LSTUR [1], NPA [43], and NRMS [44].<sup>6</sup> Recommendations are calculated using the test set with 15,339 users. To optimize for diversity, the output of the neural models is re-ranked using three customized approaches: greedy KL (G-KL) [33], PM-2 [12], and MMR [8].<sup>7</sup> Additionally,  $RP_{\beta}^3$ , random walks with erasure (RWE-D), and a randomized list (Random) were added as baseline algorithms.<sup>8</sup>

As recommended in [16, 38], we create large enough recommendation lists for meaningful diversity, calculating and assessing the diversity metrics on the top 20 items. Drawing on the theoretical framework of Helberger [19], Vrijenhoek et al. [38] proposed five normative metrics to measure the diversity of NRSs, called the RADio metrics. The RADio metrics included in our analysis are: activation (Activ.), category calibration (Cat. Calib.) and complexity calibration (Comp. Calib.), fragmentation (Frag.), alternative voices (Alt. Voices), and representation (Repr.). We report on the traditional, non-normative diversity metrics that focus on *equality* using the Gini coefficient [9] of article category (Cat. Gini), sentiment (Sent.

Gini), and party (Party Gini) and the *intra-list distance* (ILD) [7] of category (Cat. ILD), sentiment (Sent. ILD), and party (Party ILD). Additionally, we list estimated energy cost to train models/create graphs (Train Cost) and to calculate the recommendations (Rec. Cost) in watt seconds.<sup>9</sup> Finally, we assess the model performance using the area under the curve (AUC) [30] for test impressions.<sup>10</sup>

For our NTD, we consider sentiment and political party distributions in this experiment. Sentiment ranges from negative (−1.0) to positive (1.0).<sup>11</sup> The articles are put into 4 discretized buckets with value ranges of [−1, −0.5) (20%), [−0.5, 0) (30%), [0, 0.5) (30%), [0.5, 1] (20%).<sup>12</sup> The political party is based on mentions.<sup>13</sup> NTD has buckets for articles that mention 1) government (incl. supporting parties, 15%), 2) opposition parties (15%), 3) government *and* opposition parties (15%), 4) independent/foreign parties (*must* mention at least one independent/foreign party, *may* mention a majority party, 15%), and 5) no political parties (40%). The evaluation is done with an extended version of the Cornac [32, 34, 35].

## 5 Results

Table 1 shows the results of the RADio diversity, traditional diversity metrics of equality and distance, cost estimates, and AUC. Gini and ILD for NTD (Perfect) across sentiment and political party are optimal values.

**Activation** compares article sentiment distributions, with higher values indicating a larger divergence between a user’s recommendation list and the item pool. D-RDW and MMR for NPA and NRMS result in the largest divergence.

<sup>6</sup>We calculate the energy cost based on the power consumption (in watts for CPU and GPU components) and factor in the overall runtime [37].

<sup>10</sup>The AUC score is unaffected by re-ranking and thus skipped in our reporting.

<sup>11</sup>We used the XLM-roBERTa model: <https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base-sentiment>

<sup>12</sup>Percentages indicate the share of each bucket in the final recommendation list (e.g., 20% of all articles in the user’s feed should have a positive sentiment of 0.5 or higher).

<sup>13</sup>EB-NeRD covers predominantly news from Denmark. We therefore model NTD according to the Danish political landscape, with governing and supporting parties as the majority and the opposition parties as the minority. Wikipedia overview: <https://en.wikipedia.org/wiki/Folketing>

<sup>6</sup>For each model, we use the hyperparameters from the original papers and copied the implementation found on the Microsoft Recommenders GitHub repository: <https://github.com/recommenders-team/recommenders>

<sup>7</sup>G-KL, PM-2, and MMR use the same NTD sampler as D-RDW. MMR creates an equal distribution across the target dimensions.

<sup>8</sup>For augmenting the graphs of the random walk models, we connected each cold item with the users of their 3 most similar items.

**Calibration** compares the category and complexity distribution in the user history and their recommendation list. A higher value signifies a greater deviation from the user’s category/complexity preferences. D-RDW,  $RP_\beta^3$ , and LSTUR (with and without MMR) show the smallest divergence for category; RWE-D, D-RDW, and  $RP_\beta^3$  have the smallest divergence for complexity.

**Fragmentation** quantifies the difference in story distribution across users. A higher value indicates a greater variation in story chains, showing a more fragmented user base. D-RDW, RWE-D, and NPA with PM-2 provide the least fragmented user base.

**Alternative Voices** compares the proportion of minority and majority perspectives within the recommendations. A higher value indicates a greater disparity between the two perspectives. The random baseline, NRMS, and NPA show the smallest disparity.

**Representation** measures the divergence in representing political parties. A higher value indicates a larger difference between the recommendation list and the item pool. LSTUR and NPA with MMR, NRMS with MMR, and D-RDW show the largest divergence.

**Gini coefficients** quantify inequality in the target dimension. A higher value indicates larger inequality. D-RDW and the G-KL re-ranker outperform the base neural models across Gini of sentiment and party (with perfect scores for the NTD dimension). The random baseline achieved the best results for category Gini.

**ILD** calculates the average pairwise dissimilarity between items using cosine distance. A higher value means more dissimilarity (i.e., diversity) between items. D-RDW and the G-KL re-ranker have the lowest ILD for the dimensions of sentiment and party, with the random baseline performing best on category ILD.

**Computational cost** lists the average watt seconds for calculating the recommendation list of 20 items per user.<sup>14</sup> We see that the random walk models require two orders of magnitude less energy for training compared to the neural models and one to almost three orders of magnitude less for the recommendations process.

**AUC** assesses model classification quality in predicting item clicks, with 0.5 being the expected value of random guessing. We see  $RP_\beta^3$  has the highest AUC, followed by LSTUR in the second place, and a tie between D-RDW, RWE-D, and NPA in third place.

## 6 Discussion

There are four dimensions to our performance comparison: 1) normative RADio diversity,<sup>15</sup> 2) traditional Gini/ILD diversity, 3) energy cost, and 4) AUC. The results in Table 1 show that D-RDW has consistent scores in the top 3 across all four of these dimensions. Furthermore, the NTD re-ranking approaches outperform the original neural models on non-normative diversity.<sup>16</sup>

<sup>14</sup>The training phase for the random walk models is the graph construction (incl. graph augmentation). The recommendation phase for graph-based algorithms includes random walks, applying the distribution, and item re-ranking.

<sup>15</sup>Interpreting RADio is challenging. It expresses a divergence. Hence, if the dataset is diverse, a small divergence is desirable and the other way around. Therefore, we need to first know the effective NTD-relevant distributions in EB-NeRD. For political party mentions, we have 2.66% for the governing parties and their supporters, 0.88% for the opposition parties, 0.41% for both governing and opposition parties, 16.96% for independent and foreign parties, and 79.09% of articles have no political entities. In terms of sentiment, 9.51% are negative, 58.01% somewhat negative, 30.01% somewhat positive, and 2.48% positive.

<sup>16</sup>We share additional results in our online repository with values using the Microsoft News Dataset (MIND) [46] to show that the performance of D-RDW is consistent across different datasets.

Looking at RADio, we see that random walk models show the best calibration for user preferences and minimize overall fragmentation. For traditional diversity, the NTD in-process resampling of D-RDW and the NTD post-processing for re-rankers allow targeting and boosting of specific metrics to achieve perfect scores across Gini as well as ILD for sentiment and party. As such, applying NTD to the recommender pipeline is an effective way for editors and journalists to boost specific Gini and ILD scores.

While neural models with and without NTD re-ranking allow for combining AUC gains with excellent Gini and ILD performance, it comes at around 400-1000 times the cost of random walk algorithms. Even if one disregards ecological concerns and diversity optimization, there is an economic argument to be made to use random walks. They allow for cheaper/more frequent recommendations, with cheaper updates [28], which is of particular importance in the news domain where items have a relatively short shelf life.

Lastly, we see that the random walk model  $RP_\beta^3$  can outperform AUC-optimized neural models on EB-NeRD, with D-RDW and RWE-D achieving scores comparable with LSTUR and NPA. While AUC values below 0.6 indicate poor discriminatory power, it is important to emphasize that the news presents a difficult domain. Reference AUC values for state-of-the-art algorithms are around this value for MIND [46], EB-NeRD [24],<sup>17</sup> and NeMig [21].

## 7 Limitations and Future Work

Since the dataset heavily influences the performance of recommender models and re-rankers, future work should better focus on examining the relationship between dataset characteristics and diversity outcomes. Second, further experiments could focus on better understanding the normative target distributions for multiple diversity dimensions on user engagement and satisfaction. Third, integrating personalized approaches to diversity is a promising direction for future work (i.e., what level of diversity is suitable for what type of user). To facilitate this, a user modeling module could be developed to enhance the system’s ability to understand and adapt to user attributes, such as preferences for news topics, political orientation, or desired levels of diversity. Lastly, user studies are required to understand what users perceive/accept as diverse and satisfying [16, 39].

## 8 Conclusion

In this paper, we present D-RDW, a first operationalization of deliberative recommenders from the normative framework of Helberger [19]. It is the best-performing solution when looking at the combined dimensions of RADio, Gini/ILD, energy cost, and AUC when compared to state-of-the-art neural models and random walks. Furthermore, we provide a stand-alone version of the NTD sampler of D-RDW as a re-ranker for existing neural models. The re-rankers allow for achieving perfect results when added to the neural models across the traditional ILD and Gini diversity metrics. This enables the input of editorial and journalistic values recommendations. We hope D-RDW and random walks in general can become a cornerstone of exploring norm-aware NRSs that go beyond the accuracy objective to assess and consider the societal impact.

<sup>17</sup>Due to data leakage, only the subset of algorithm using the original dataset are comparable to our D-RDW: <https://www.codabench.org/forums/2387/342>

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